



Parul University
Faculty of Engineering and Technology
Parul Institute of Engineering and Technology
Department of Artificial Intelligence and Data Science

Subject Name	Theory of Computations	A.Y	2026-27
Subject Code	303105306	Semester	5

Assignment-4

Sr No	Question	COs	B.T	Competence
1	Define a Turing Machine (TM) and explain its basic components with a suitable block diagram and example.	1	1	Remember
2	Explain the difference between Turing-recognizable (Recursively Enumerable) languages and Turing-decidable (Recursive) languages with suitable examples.	4	2	Understand
3	Design a Turing Machine that accepts the language $L = \{a^n b^n \mid n \geq 1\}$. Demonstrate the working of the machine for the string "aabb".	4	3	Apply
4	Analyze the differences between Deterministic Turing Machines and Nondeterministic Turing Machines. Discuss their equivalence in computational power with suitable examples.	4	4	Analyze
5	Evaluate the closure properties of Recursive and Recursively Enumerable languages under union, intersection, and complement operations. Justify your answer with proper reasoning.	5	5	Evaluate
6	Construct an unrestricted grammar equivalent to a given Turing Machine or design a TM Enumerator for a suitable language. Explain the equivalence between unrestricted grammars and Turing Machines.	4, 5	6	Create